

ERICA CHIANG

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EDUCATION

Cornell University

Ph.D. candidate in Computer Science

Aug 2023 - present

Carnegie Mellon University

B.S. in Computer Science, Minor in Human-Computer Interaction

Cumulative GPA: 3.95 / 4.0

Phi Beta Kappa, University Honors, School of Computer Science College Honors

Aug 2019 - May 2023

AWARDS

Finalist , Michael H. Rothkopf Junior Researcher Paper Prize (winners TBA)	2026
Finalist , Doing Good with Good OR Student Paper Competition (winners TBA)	2026
Best Paper with Student Lead Author , EC 2026	2026
Best Poster Award , IOE-ISyE-MS&E Rising Stars Workshop	2026
Rising Star in IOE-ISyE-MS&E , University of Michigan Industrial and Operations Engineering	2026
Best Paper Award , CHIL 2025	2025
Digital Life Initiative Doctoral Fellowship	2025
Siegel PiTech PhD Impact Fellowship	2025
NSF Graduate Research Fellowship	2023
Phi Beta Kappa	2023
Carnegie Mellon Senior Leadership Recognition Award	2023
Andrew Carnegie Society Scholar <i>40 out of 2000 students selected for academic excellence and leadership</i>	2023
Undergraduate Student Research Competition 2nd Place , ACM SIGCOMM	2022
Johns Hopkins Applied Physics Laboratory Positive Influence Award <i>1 out of 200 interns selected for exceptional performance and leadership</i>	2022

PUBLICATIONS

- The Forest from the Trees: Lessons on Public-Sector Deployment from an Academic Collaboration**
Erica Chiang, Katelyn Liu, Anum Ahmad, Zhi Liu, Gabriel Agostini, Tyler Gibson, Uma Bhandaram, Nikhil Garg
In submission
- Personalized Recommendations Without Inducing Congestion: Mitigating Disparities in the NYC High School Match**
Erica Chiang, Kenny Peng, Rebecca Lichtenstein, Brielle McDaniel, Kristen O'Neil, Dejaunique Thomas, Lianna Wright, Jon Kleinberg, Eva Tardos, Nikhil Garg
ACM Conference on Economics and Computation, 2026 (EC 26) 🏆 *Best Paper with Student Lead Author*
International Conference on Computational Social Science, 2026 (IC2S2 26) 🏆 *Plenary Talk (18 of 784 submissions)*
🏆 *INFORMS Michael H. Rothkopf Junior Researcher Paper Prize finalist (winners TBA) 2026*
🏆 *INFORMS Doing Good with Good OR Student Paper Award finalist (winners TBA) 2026*
🏆 *Best Poster Award at IOE-ISyE-MS&E Rising Stars Workshop 2026*

3. **Learning Disease Progression Models That Capture Health Disparities**

Erica Chiang, Divya Shanmugam, Ashley Beecy, Gabriel Sayer, Deborah Estrin, Nikhil Garg, Emma Pierson
Conference on Health, Inference, and Learning, 2025 (CHIL 25)
Machine Learning for Health Symposium, 2024 (ML4H 24)
ICML Workshop on Structured Probabilistic Inference and Generative Modeling, 2024 (SPIGM 24)
 *Best Paper Award at CHIL 2025*

4. **SurgeProtector: Mitigating Temporal Algorithmic Complexity Attacks using Adversarial Scheduling**

Nirav Atre, Hugo Sadok, **Erica Chiang**, Weina Wang, and Justine Sherry
ACM Special Interest Group on Data Communication, 2022 (SIGCOMM 22)

SELECTED TALKS

1. **Personalized Recommendations Without Inducing Congestion: Mitigating Disparities in the NYC High School Match**

- IC2S2 Plenary, Summer 2026
- Workshop on Algorithms for Learning and Economics (WALE), Summer 2026
- EC Plenary, Summer 2026
- ESIF-AIML Meeting, Summer 2026
- Market Innovation Workshop, Spring 2026
- Guest lecture, Practical Applications in Machine Learning (INFO 5368) @ Cornell Tech, Spring 2026
- Guest lecture, Networks and Markets (CS 5854 / ORIE 5138) @ Cornell Tech, Spring 2026
- EAAMO Urban Data Working Group, Spring 2026
- Cornell Tech Digital Life Initiative Seminar, Spring 2026
- Guest lecture, Applied Data Science (ORIE 5355) @ Cornell Tech, Fall 2025
- Institute for Operations Research and the Management Sciences (INFORMS), Fall 2025

2. **The New York City High School Match**

- Guest lecture, Interacting with AI: Understanding the Relationship Between AI and Humans @ Cornell Tech Summer Innovation Intensives (high school students), Summer 2026

3. **Behind the Scenes on “Personalized Recommendations Without Inducing Congestion”**

- EC 2026 workshop “From theory to practice: Behind the scenes on research deployments”, Summer 2026

SELECTED POSTERS

1. **Robust Heuristics: Attacks and Defenses on Job Size Estimation for WSJF Systems**

Erica Chiang, Nirav Atre, Hugo Sadok, Weina Wang, and Justine Sherry
ACM SIGCOMM Poster Session, 2022

 *Runner-up of the ACM SIGCOMM 2022 Undergraduate Student Research Competition*

WORK EXPERIENCE

Johns Hopkins Applied Physics Laboratory, Computer Science Intern Summer 2022

Contributed to 2 projects. (1) Chief Digital and Artificial Intelligence Office aircraft readiness model: testing, evaluation, and benchmark model development; (2) Airborne Collision Avoidance System (ACAS): designed and deployed a full stack web application for use in the development of ACAS software.

NASA Jet Propulsion Laboratory, Software Engineer Intern Summer 2021

Developed software to monitor telemetry data traveling through NASA Deep Space Network and created an interactive webpage for visualizing and accessing the data in real time, intended for real world use on vehicles such as NASA Mars Perseverance Rover.

PREVIOUS RESEARCH EXPERIENCE

Characterizing Social Media Narratives, CMU CASOS Lab Sept 2022 - May 2023

Senior Thesis Project, Advised by Kathleen Carley

Analyzed Twitter data to identify patterns in the linguistic and psychological cues that different actors tend to use, in order to understand how this shapes their influence over online communities

Adversarial Job Scheduling in Network Functions, CMU SNAP Lab

Jan 2021 - Aug 2022

Advised by Justine Sherry and Weina Wang

*Studied packet-scheduling algorithms that mitigate the damage of algorithmic complexity attacks (ACA) and led a project to explore the theoretical implications of using heuristics for packet job size estimation***Bias in ML Algorithms, CMU CHIMPS Lab**

Aug 2020 - Dec 2020

Advised by Jason Hong, Motahhare Eslami, Ken Holstein, Hong Shen

*Developed research questions and interview protocols to study how people search for, identify, and respond to bias in machine learning algorithms, with the goal of creating a crowd audit platform***TEACHING EXPERIENCE****PiTech Impact Studio** (TECH 5930 @ Cornell Tech), Teaching Assistant Spring 2026**Data Science in the Wild** (CS 5304 @ Cornell Tech), Teaching Assistant Spring 2025**Mathematical Foundations for Computer Science** (15-151/21-128 @ CMU), Teaching Assistant Fall 2021*Taught two recitation sections per week, held weekly Office Hours, designed exercises, prepared & taught exam review sessions (to 200 students each time)***MENTORING & SERVICE****Reviewer**, EconML Workshop @ NeurIPS 2026**Reviewer**, AAAI special track on Artificial Intelligence for Social Impact (AAAI AISI) 2026**Reviewer**, Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) 2026**Reviewer**, International Conference on Computational Social Science (IC2S2) 2026**Social Chair**, PhDs at Cornell Tech 2025 - 2026**Visit Day Czar**, Cornell CS Department 2025**Reviewer**, ACM Web Conference 2025**Mentor**, Cornell CS PhD Application Support Program Fall 2023, Fall 2024**Mentor**, CMU Society of Women Engineers & SCS Mentorship Programs 2022 - 2023**Orientation Counselor**, CMU First-Year Orientation 2021**LEADERSHIP / EXTRACURRICULAR INVOLVEMENTS****CMU C# Choir** 2019 - 2023

President (2021-22)

Vice President Internal (2020-21)

Design Chair (2019-20, 2022-23)

CMU Sweepstakes ("Buggy") 2019 - 2023*Women's Push Captain (2022-23): Recruit and lead team of runners to compete in annual Spring Carnival***CMU Taiwanese Students Association** 2019 - 2023

Secretary (2021-22)

Freshman Representative (2019-20)

CMU Club Soccer Team & Lake Oswego High School Varsity Soccer 2015 - 2021**Lake Oswego High School Student Government** 2018 - 2019

Student Body Vice-President (2018-19)

Lake Oswego High School Varsity & Club Track and Field 2015 - 2019

Team Captain (2019)

*Oregon Class 6A (largest division) All-State First Team x2**High school record holder*

Physics Lab Educator, Oregon Museum of Science & Industry

2016 - 2019

Developed new lab demonstrations and experiments for museum visitors, worked with children to teach physics concepts through visual and interactive activities

SKILLS

Languages: C, Python, Java, HTML, JavaScript, CSS, MATLAB

Art and Design: ericchiang.wixsite.com/website-1

National Scholastic Art Awards 2019: 1 Gold Key, 2 Silver Keys